Winning Isn't Everything: Corruption in Sumo Wrestling

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Duggan & Levitt (2002) · American Economic Review 92(5): 1594–1605

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I. Background & Research Gap

Why is corruption hard to study — and why does it matter?

The Core Problem: Corruption Hides Its Tracks

- **Corruption is generally agreed to be widespread and harmful**
  - Reduces economic efficiency (Shleifer & Vishny, 1993)
  - Distorts public procurement, lowers growth (Mauro, 1995)
- **Yet rigorous empirical evidence is scarce**
  - Corrupt actors deliberately avoid leaving a trail — no public record of bribes or agreements
  - Standard datasets will almost never contain direct evidence of illicit behavior
- What exists? Mostly perception indices and anecdote
  - Porter & Zona (1993): bid-rigging in highway auctions
  - Di Tella & Schargrodsky (2000): prices at Buenos Aires hospitals fell 10–20 % after a corruption crackdown
  - Fisman (2001): Indonesian firms connected to Suharto fell more on news of his ill-health
- **D&L's insight: incentives create predictable behavioral fingerprints — even without direct evidence**
  - Find a setting where (i) incentives are transparent and (ii) outcomes are observable at scale

How to Find Corruption

There are two types of evidence of corruption:

① Direct Evidence

- E-mails & phone calls
- Whistleblowers
- Financial records / payments
- Rarely available: corruptors hide communication

② Behavioral Methods

- Statistical anomalies in observed outcomes
- Patterns predicted by incentive theory
- Cannot hide behavior — only the act itself
- Does not distinguish explicit vs. tacit collusion

Key insight: Corrupt parties can hide communication — but they cannot hide the *effect* of their behavior on observable outcomes. Statistical screens exploit this.

- This paper: derive predictions from the incentive structure → test with 11 years of match data
- Small note on Vocabulary: Corruption and collusion are not strictly defined (in this context) and can be used interchangeably. However, collusion is more common in market design and auction theory. Corruption on the other hand is more frequently found in development economics and is often referring to the government or a bureaucracy.

Source: Intro slides, Prof. Graeber (Feb 24, 2026); Duggan & Levitt (2002)

II. Why Sumo?

An unlikely but ideal laboratory

Why Sumo Wrestling?

- **If corruption thrives here, no institution is safe**
  - Sumo is Japan's national sport — 2,000 years of honor, ritual, and tradition
  - Japan consistently ranks among the world's least corrupt countries (Transparency International, 2000)
- **Exceptionally transparent institutional framework**
  - Simple, fixed rules — easy to model incentives and derive falsifiable predictions
  - Promotion, demotion, salary schedules, and tournament format are all publicly documented
- **Ideal data environment**
  - Every official match result publicly recorded; large balanced panel with repeated pairings
  - 281 wrestlers, around 64,000 wrestler-matches, Jan 1989 – Jan 2000
- **Long-running allegations — never substantiated**
  - Insiders have alleged match-fixing for years; the Japan Sumo Association dismisses all claims
  - Despite years of allegations, zero formal disciplinary actions taken as of 2002
- **The method generalizes: sumo is a proof of concept for forensic economics**

III. The Critical 8th Win

A sharp nonlinearity in the payoff function

Tournament Format

- **Six tournaments (basho) per year**
  - Each tournament lasts 15 days; each wrestler fights exactly once per day
  - Top division (Sekitori): ~66 wrestlers (rikishi)
- A fight is won when the opponent is pushed out of the ring or touches the ground first
  - Tournament won by most wins; ties resolved by play-off
- **After each tournament: rankings (banzuke) are adjusted**
  - **≥ 8 wins → kachi-koshi → rank rises**
  - **≤ 7 wins → make-koshi → rank falls**
- **Rank is real currency — not just symbolic**
  - Salary, prestige, and daily privileges all depend on rank
  - Low-ranked wrestlers do chores; reaching juryo eliminates those duties
  - Top-40 wrestlers (maegashira +) have their own attendants
  - **Rule of thumb: 1 ranking spot ≈ $3,000/year in additional earnings**

The Payoff Nonlinearity — Figure 1

- Nearly linear relationship between wins and rank change — except win #8
  - Wins 1–7 and 9–11: each extra win ≈ +3 rank spots
  - **Win #8: ≈ +11 rank spots (≈ 4× the typical value)**
- **Nonlinear Payoff**
  - 7–7 wrestler: marginal gain of ≈ 11 ranks from winning
  - 8–6 opponent: loses only ≈ 3 ranks if he drops the match
- **Monetary stakes**
  - 8th win ≈ 11 spots × $3,000 / year ≈ $33,000 in annual income
  - No legally binding contract can be written — enforcement must come from repetition and social ties

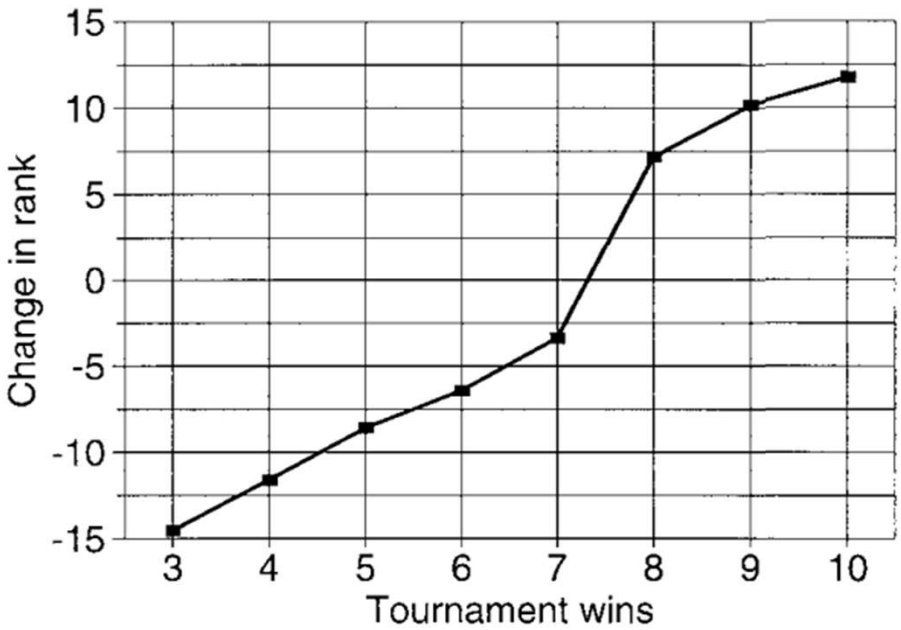

FIGURE 1. PAYOFF TO TOURNAMENT WINS

IV. Data & Empirical Strategy

What do we observe — and how do we test?

Data

- **Source & coverage**
  - All official sumo matches, January 1989 – January 2000
  - Top division (Sekitori) only; 6 tournaments per year × ~11 years
- **Scale**
  - 281 wrestlers; over 64,000 wrestler-matches (≈ 32,000 bouts)
  - <5% discarded due to missing data, coding errors, or early injury withdrawal
  - Average of 10 meetings between any two wrestlers
- **Variables per match**
  - Wrestler identities, outcome (W/L), tournament month/year, day of match (1–15), stable affiliation (≈ 98% coverage)
- **Defining 'bubble' matches**
  - Day 15: wrestler has exactly 7 wins (needs exactly 1 more for kachi-koshi)
  - Day 14: wrestler has 6 or 7 wins; Day 13: 5, 6, or 7 wins
  - **Bubble = 1 if wrestler on margin; −1 if opponent is; 0 otherwise**

Empirical Strategy — Regression Model

Linear probability model (OLS):

$$Win_{ijtd} = \beta \mathbf{Bubble_{ijtd}} + \gamma \mathrm{Rankdiff}_{ijt} + \lambda_{ij} + \delta_{it} + \epsilon_{ijtd}$$

- **Variables**
  - Win_{ijtd} = 1 if wrestler i beats opponent j on day d of tournament t
  - $Bubble_{ijtd}$ = +1 if only wrestler i is on the margin; −1 if only opponent is; 0 otherwise
  - $\mathrm{Rankdiff}_{ijt}$ = official rank of i minus rank of j (controls for ability gap)
- **Identification**
  - Tested with and without wrestler / opponent fixed effects (λ_{ij})
  - Also tested with wrestler–opponent interaction effects (within-pair variation only)
- Standard errors corrected for two observations per bout (one per wrestler)
- **The coefficient β captures excess win probability when on the bubble, controlling for ability**

V. Evidence of Match Rigging

What do the data actually show?

Evidence I: The Win Distribution

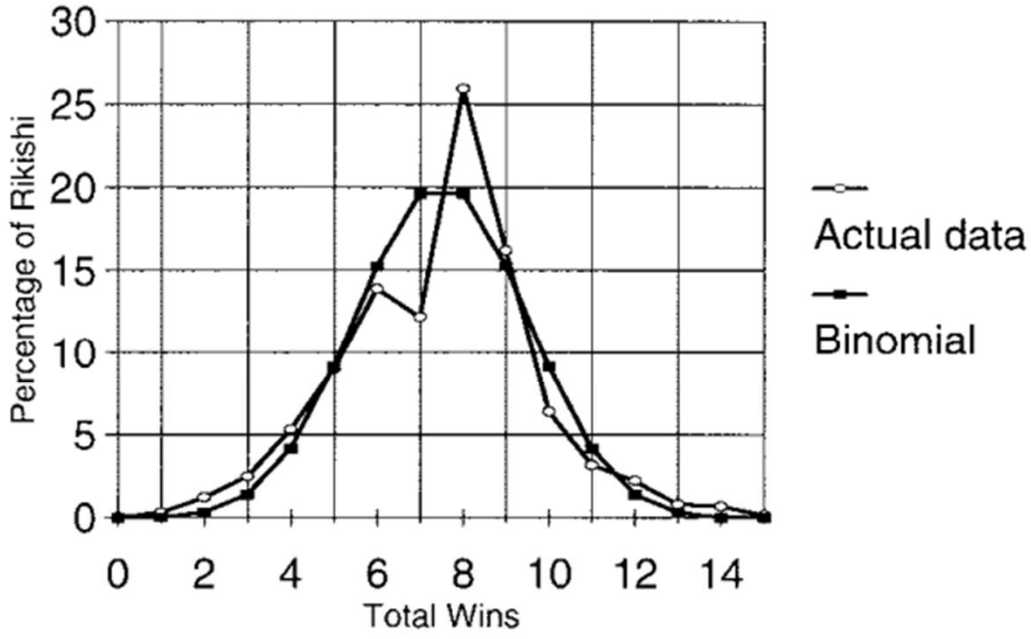

FIGURE 2. WINS IN A SUMO TOURNAMENT (ACTUAL VS. BINOMIAL)

- **If all matches were fair (coin toss):**
  - 7 wins and 8 wins should each occur with frequency 19.6%
- **What we observe:**
  - **8 wins: 26.0% (+6.4 pp above expected)**
  - **7 wins: 12.2% (−7.4 pp below expected)**

Evidence II: Excess Win Probabilities by Day (Table 1)

TABLE 1—EXCESS WIN PERCENTAGES FOR WRESTLERS ON THE MARGIN FOR ACHIEVING AN EIGHTH WIN, BY DAY OF THE MATCH

On the Margin on:	(1)	(2)	(3)	(4)	(5)	(6)
Day 15	0.244	0.249	0.249	0.255	0.260	0.264
	(0.019)	(0.019)	(0.018)	(0.019)	(0.022)	(0.022)
Day 14	0.150	0.155	0.152	0.157	0.168	0.171
	(0.016)	(0.016)	(0.016)	(0.016)	(0.019)	(0.019)
Day 13	0.096	0.107	0.110	0.118	0.116	0.125
	(0.016)	(0.016)	(0.016)	(0.016)	(0.019)	(0.019)
Day 12	0.038	0.061	0.064	0.082	0.073	0.076
	(0.017)	(0.018)	(0.017)	(0.018)	(0.020)	(0.021)
Day 11	0.000	0.018	0.015	0.025	0.010	0.012
	(0.018)	(0.018)	(0.018)	(0.018)	(0.021)	(0.021)
Rank difference	—	0.0053	—	0.0020	—	−0.0020
		(0.0003)		(0.0003)		(0.0004)
Constant	0.500	0.500	—	—	—	—
	(0.000)	(0.000)				
R^2	0.008	0.018	0.030	0.031	0.0634	0.0653
Number of observations	64,272	62,708	64,272	62,708	64,272	62,708
Wrestler and opponent fixed effects	No	No	Yes	Yes	Yes	Yes
Wrestler-opponent interactions	No	No	No	No	Yes	Yes

VI. Effort vs. Corruption

Can the pattern be explained without collusion?

The Effort Hypothesis

7-7 Wrestlers have more to win, and to lose, so they try harder.

Test 1: Media scrutiny

May 1996, Nov 1999, Jan 2000 were periods of intense press coverage of match-rigging allegations.

Bubble win rates return completely to baseline under scrutiny.

Effort does not fall when more people watch. Collusion does.

Test 2: Special prizes

When the opponent is in the running for a special prize (Outstanding Performance, Technique, Fighting Spirit), the bubble advantage disappears completely.

An opponent with high own stakes refuses to throw the match.

Test 3: Career horizon

Bubble success rises over a wrestler's career — consistent with reputation building. But in the final year of a career, it falls again.

'Shadow of the future': near retirement, future favors are worthless. Effort has no reason to change.

Test 4: Years of experience

A 5-year veteran is ~4 pp more likely to win on the bubble than a rookie — all else equal.

Consistent with collusive reputation taking time to build, not with physical peak performance.

Evidence III: Future Match Reversal (Table 3)

If a wrestler wins because his opponent threw the match, the opponent must be compensated.

- **Mechanism: payment in kind**
  - Cash leaves a paper trail
  - A future win is 'inside the arena' is harder to prove
- **Prediction**
  - After winning on the bubble, wrestler should lose more often vs. the same opponent next time
  - But only for winners, losers on the bubble show no reversal
- **Result**
  - **Winners on the bubble: ~8 pp more likely to lose the next meeting**
  - Losers on the bubble: no significant reversal (−5.6 pp, n.s.)
- **~⅔ of all excess bubble wins are returned in the very next meeting**
- Suggests cash payments also exist — one future loss is too cheap to be the whole compensation

TABLE 3—WIN PERCENTAGES IN PRECEDING AND SUBSEQUENT MATCHES
(For Two Wrestlers Who Meet When One is on the Margin in the Final Three Days of a Tournament)

	All Matches on the Margin		Only Matches in Which the Wrestler on the Margin Wins		Only Matches in Which the Wrestler on the Margin Loses	
Variable	(1)	(2)	(3)	(4)	(5)	(6)
One or two matches prior to the bubble match	−0.002 (0.009)	0.005 (0.012)	0.020 (0.011)	0.019 (0.017)	−0.041 (0.016)	−0.035 (0.022)
Bubble match	0.151 (0.010)	0.164 (0.014)	—	—	—	—
First meeting after bubble match	−0.073 (0.011)	−0.062 (0.015)	−0.082 (0.015)	−0.079 (0.020)	−0.056 (0.020)	−0.040 (0.027)
Second meeting after bubble match	−0.002 (0.013)	0.005 (0.016)	0.031 (0.017)	0.028 (0.022)	−0.061 (0.023)	−0.039 (0.030)
Three or more meetings after bubble match	−0.010 (0.006)	0.012 (0.011)	0.013 (0.007)	0.022 (0.014)	−0.045 (0.008)	−0.013 (0.017)
Constant	0.500 (0.000)	—	0.500 (0.000)	—	0.500 (0.000)	—
Wrestler-opponent interactions?	No	Yes	No	Yes	No	Yes
R^2	0.008	0.271	0.002	0.279	0.002	0.279

Asymmetry: Only winners on the bubble show the subsequent reversal — exactly what a repayment mechanism predicts. Effort cannot explain this.

Evidence III: Future Match Reversal (Table 3)

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Three or more meetings after bubble match	−0.010 (0.006)	0.012 (0.011)	0.013 (0.007)	0.022 (0.014)	−0.045 (0.008)	−0.013 (0.017)
Constant	0.500 (0.000)	—	0.500 (0.000)	—	0.500 (0.000)	—
Wrestler-opponent interactions?	No	Yes	No	Yes	No	Yes
R^2	0.008	0.271	0.002	0.279	0.002	0.279

Evidence IV: Stable-Level Collusion (Table 2)

- **Structure of sumo stables**
  - Each wrestler belongs to a stable (heya) for their entire active career
  - Stable-masters exert strong influence over career and daily life
  - Wrestlers from the same stable never meet in tournaments — but inter-stable matches are common
- **Hypothesis: reciprocity agreements between stables**
  - If stable A and stable B have a deal, A's wrestlers do well on the bubble vs. B — and vice versa
  - Test: correlate historical cross-stable bubble win rates with today's outcome (excluding current match)
- **Result**
  - **For every 10 pp higher historical cross-stable bubble success rate: +3 pp today (highly significant)**
  - No plausible effort story: if effort drove results, stables good at effort should also be good at defending against motivated opponents → predicts zero or negative correlation
- **Interpretation: match fixing is not purely bilateral — it operates as network collusion coordinated through stable hierarchies**
  - This increases enforceability: sanctions can be applied at the stable level, not just to individuals

Evidence IV: Stable-Level Collusion (Table 2)

TABLE 2—DETERMINANTS OF EXCESS WIN LIKELIHOODS FOR WRESTLERS ON THE BUBBLE

Variable	(1)	(2)	(3)
Wrestler on bubble	0.126	0.117	0.155
	(0.026)	(0.026)	(0.029)
Wrestler on bubble interacted with:			
High media scrutiny	−0.188	−0.177	−0.146
	(0.071)	(0.071)	(0.080)
Opponent in running for a prize this tournament	−0.149	−0.129	−0.156
	(0.047)	(0.046)	(0.052)
Number of meetings between two opponents in the last year	−0.0048	−0.0031	−0.0024
	(0.0082)	(0.0081)	(0.0096)
Wrestler on bubble in his last year of competing	−0.0361	−0.0195	−0.0346
	(0.0398)	(0.0395)	(0.0493)
Years in sumo for wrestler on bubble	0.0077	0.0077	0.0091
	(0.0036)	(0.0036)	(0.0043)
Winning percentage in other bubble matches between these two stables	0.272	0.293	—
	(0.059)	(0.058)	
R^2	0.016	0.074	0.246
Wrestler and opponent fixed effects?	No	Yes	Yes
Wrestler-opponent interactions?	No	No	Yes

VII. Whistleblower Corroboration

Does insider knowledge match the data?

Insider Allegations Meet Data (Table 4)

- Two former wrestlers publicly named 29 wrestlers as corrupt and 14 as clean
- D&L test: do the named 'corrupt' wrestlers outperform on the bubble vs. named 'clean' wrestlers?

Opponent →	Corrupt	Status unknown	Clean
Corrupt wrestler on bubble	**+26.0 pp*** **	**+27.1 pp*** **	+3.6 pp n.s.
Status unknown on bubble	**+27.0 pp*** **	**+18.1 pp*** **	+4.1 pp n.s.
Clean wrestler on bubble	+3.6 pp n.s.	−3.3 pp n.s.	+2.2 pp n.s.

- **Corrupt vs. corrupt or unknown: excess win rates of +18 to +27 pp — strongly significant**
- **Any match involving a clean wrestler: no significant deviation from expected outcomes — in any configuration**
- The insider allegations are directly confirmed by the data — and almost all 'unknown' wrestlers also appear to be corrupt

Insider Allegations Meet Data (Table 4)

TABLE 4—EXCESS WIN PERCENTAGES ON THE BUBBLE FOR WRESTLERS LABELED BY SUMO INSIDERS AS "CORRUPT" OR "CLEAN"

		Wrestler on the Bubble Is Identified as:		
		Corrupt	Status Unknown	Clean
Opponent of wrestler on the bubble is identified as:	Corrupt	0.260	0.270	−0.010
		(0.037)	(0.021)	(0.038)
	Status unknown	0.271	0.181	0.041
		(0.021)	(0.019)	(0.031)
	Clean	0.036	−0.033	0.022
		(0.027)	(0.035)	(0.074)

VII. Our Assessment

What does this teach us about incentives and corruption?

Our Assessment of the Paper

What we find compelling:

- The bundle of evidence: no single test is decisive, but all tests point in the same direction
  - The rematch reversal is a convincing argument against the effort hypothesis
  - The whistleblower confirmation is remarkably precise: the named wrestlers deviate, the 'clean' ones do not

Notes on external validity:

- The method can only detect one type of collusion, agreements between two wrestlers or stables, and only if the payment is an In-kind transfer
- Screening for collusion is always case specific
- Changing the incentive structure is not a solution to collusion in most markets

Policy Implications

① Redesign the payoff structure

The 8th-win nonlinearity is the economic root of match rigging. Smoothing the payoff function reduces the gains from trade. The Japan Sumo Association effectively proved this: when they lowered the value of the 8th win (2000–2003), the suspicious patterns weakened. Policy lesson: sharp discontinuities in rank, bonus, or promotion rules invite manipulation.

② Increase monitoring and public scrutiny

The analysis shows that match rigging disappears during periods of intense media coverage and returns when attention fades. Detection risk is itself a deterrent. Practical implications: independent auditing, transparent outcome data.

③ Provide strong incentives for whistleblowers

Insiders have accurate knowledge: the two sumo whistleblowers named exactly the right people. When insider information can be corroborated by objective statistical analysis, it becomes much more credible and actionable.

Conclusion

Sharp incentive nonlinearity creates gains from trade between wrestlers — the 8th win is worth ≈ 4× a normal win, generating a clear motive for match fixing.

Multiple independent tests — win distribution, rematch reversal, stable reciprocity, career horizon, whistleblower corroboration — all point in the same direction.

Broader message: Incentives, information, and strategic behavior are everywhere — in daycares, gyms, sports arenas, and living rooms.

Policy: remove incentive discontinuities, increase monitoring, and incentivize whistleblowers. Corruption can be reduced or made visible through institutional design.

Duggan & Levitt (2002) · AER 92(5) | Dietl, Lang & Werner (2010) · JSE 11(4)

IX. Update: Dietl, Lang & Werner (2010)

What happened after D&L was published?

Dietl, Lang & Werner (2010) — Overview

- **Research question**
  - Does corruption in sumo persist after D&L's findings became public?
  - Does the pattern respond to changes in the incentive structure and public scrutiny?
- **Data**
  - 33,734 matches, January 1995 – November 2006 (283 wrestlers)
  - Sources: scgroup.com, Japan Sumo Association, Hungarian Sumo Association
- **Three periods (split around D&L's publication process)**
  - Period 1: Jan 1995 – Jan 2000 (before D&L presented publicly)
  - Period 2: Mar 2000 – May 2003 (high public scrutiny during & after publication)
  - Period 3: Jun 2003 – Nov 2006 (corruption reappears)
- **Main finding**
  - Strong statistical evidence that corruption is reduced in Period 2 but reappears in Period 3
  - The incentive nonlinearity also disappears in Period 2 and returns in Period 3 — consistent with the sumo association adjusting the ranking rule as a reaction to D&L

Win Distribution Across Three Periods

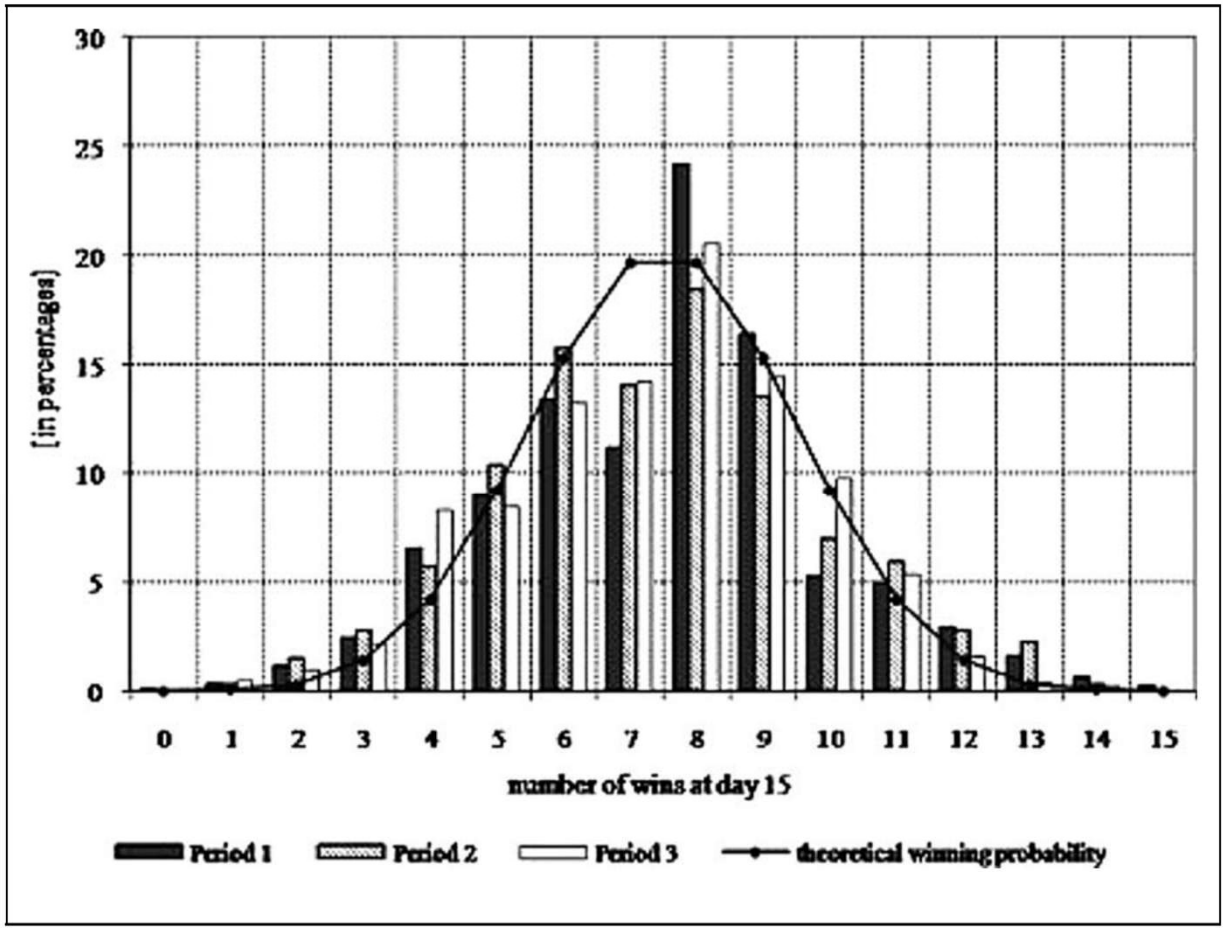

Figure 1. Frequency of Wins in a Sumo Tournament

- **Period 1: 24% finish at 8 wins (vs. 12% at 7 wins)**
- Period 2: 8-win spike shrinks to 18% — gap nearly closes
- **Period 3: 8-win spike returns to 20.5%**
- A simple binomial test shows the 7-win / 8-win asymmetry is significant in Periods 1 and 3, but not Period 2
- **Interpretation: scrutiny + incentive redesign reduced corruption; when both relaxed, corruption returned**

The Incentive Structure Across Periods

- D&L identified the 8th-win nonlinearity as the economic root of corruption
- **Dietl et al. track how that kink evolved after D&L published:**
  - **Period 1 (before publication): 8th win worth ≈ 8 rank spots (≈ 3× a typical win)**
  - **Period 2 (high scrutiny, 2000–2003): 8th win reduced to only ≈ 4 rank spots — just slightly above average**
  - **Period 3 (2003–2006): the nonlinearity returns to near-Period-1 levels**
- **Interpretation: the Japan Sumo Association responded to D&L's study**
  - By lowering the relative value of the 8th win in Period 2, the JSA removed the gains from trade — the incentive to rig vanished
  - When scrutiny faded, the nonlinearity was gradually restored, and corruption reappeared
- **This structural change is strong external validation of D&L's mechanism**
  - If the incentive kink drives corruption, removing the kink should reduce corruption — and that is exactly what happened

Update: Excess Win Probabilities Across Three Periods

Dietl et al. estimate the same regression for each period separately:

Day	Period 1	Period 2	Period 3
Day 15	**+19.8 pp*** **	+16.5 pp***	+7.4 pp n.s.
Day 14	**+17.5 pp*** **	+14.3 pp***	+11.1 pp***
Day 13	**+13.3 pp*** **	+12.6 pp***	+10.0 pp**
Day 12	+8.3 pp***	+7.2 pp n.s.	+7.9 pp*
Day 11	+0.3 pp n.s.	+4.3 pp n.s.	+5.4 pp n.s.

- Period 1 matches D&L closely: excess win probs. of +8 to +20 pp, especially in final days
- Period 2: reduction is significant for days 13–15; Day 15 excess shrinks but remains meaningful
- **Period 3: Day 15 anomaly disappears statistically, but Days 13–14 remain significant — corruption becomes more 'cautious', less end-stage-focused**

Update: Subsequent Matches & New Corruption Strategies

- **Subsequent match reversal — replicated and extended**
  - Period 1: bubble win +13.4 pp; next meeting −9.7 pp (both $p < 0.01$)
  - Period 2: both effects disappear entirely
  - Period 3: bubble win +4.9 pp; next meeting −4.2 pp (both significant)
- **New finding: 'Sudden Weakness' strategy**
  - A wrestler starts the tournament strongly (e.g. 7–1 through day 8), then loses repeatedly toward the end
  - 59 wrestlers show this win-loss pattern in ≥ 2 tournaments — all wrestlers publicly accused by insiders of being corrupt belong to this group
  - **Having a 'suddenly weak' opponent raises a bubble wrestler's win probability by ~10 pp in Periods 1 & 3**
- **New finding: Same Stable Group effect (Period 1)**
  - In Period 1, same-group matches give +9 pp additional win probability on the bubble
  - Effect disappears in Periods 2–3 — possibly replaced by more covert coordination mechanisms after publication